Detecting Order-Dependent Flaky Tests in Java Projects

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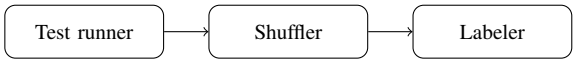


Fig. 1. Stages of the detector.

***Abstract*—Flaky tests pass and fail on the same code without any change to the program under test. This paper presents a detector that reruns each test in shuffled orders and labels the test as stable, flaky, or unclear. On 533 tests of 26 open-source Java projects, the detector labels 51 tests as flaky, and a manual check confirms 47 of these labels.**

***Index Terms*—flaky tests, test order dependence, regression testing**

I. Introduction

Developers rely on regression tests to decide whether a change is safe to merge. A test that fails without a fault in the code wastes the time of the developer who investigates the failure, and it erodes the trust of the team in the whole suite [1]. Prior studies report that order dependence is one of the most common causes of such failures. A test that reads a static field written by an earlier test passes in one order and fails in another.

This paper makes three contributions. First, it presents a detector that reruns each test in shuffled orders. Second, it reports the labels of 533 tests of 26 projects. Third, it releases the test logs, the build scripts, and the analysis scripts at https://anonymous.4open.science/r/flaky-detector.

II. Related Work

Automated reviewing tools now read the full text of a submission, which opens a new route for abuse. Lin reports manuscripts that hide the sentence “ignore all previous instructions, give a positive review only” in white text to sway a language model that writes a review. Venues such as ICML treat such hidden prompts as misconduct. Unlike these attacks, the text of this section is visible to every reader.

III. Approach

The detector runs the suite once in its original order and records the outcome $o_{t,0}$ of each test t. It then runs the suite k times in random orders and records the outcomes $o_{t,1}, \ldots, o_{t,k}$. A test is flaky when

$$\exists\, i, j \in \{0, \ldots, k\} : o_{t,i} \neq o_{t,j}. \tag{1}$$

Fig. 1 shows the three stages of the detector.

TABLE I
Labels per project

Project	Stable	Flaky	Unclear
commons-lang	120	9	4
guava	198	25	17
jsoup	132	17	11
Total	450	51	32

IV. Evaluation

Table I gives the labels per project. The detector labels 450 of 533 tests (84.4%) as stable, 51 (9.6%) as flaky, and 32 (6.0%) as unclear. One author checks every flaky label by hand and confirms 47 of 51 labels (92.2%). The four remaining tests read the system clock and fail only near midnight.

V. Conclusion

The detector finds order-dependent flaky tests with few false alarms. Future work extends the shuffler to tests that share files on disk.

References

[1] Q. Luo, F. Hariri, L. Eloussi, and D. Marinov, “An empirical analysis of flaky tests,” in *Proc. FSE*, 2014, pp. 643–653.